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Détection d'anomalies multi-domaines dans les environnements 5G à l'aide de réseaux de neurones de graphes dynamiques

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A faire en dernier :-)

Abstract

The Cloud-Edge continuum has become a central paradigm for delivering digital services, combining the scalability of centralized Cloud infrastructures with the low-latency benefits of distributed Edge resources. However, orchestrating services across this continuum is increasingly complex, as it must balance multiple and often conflicting objectives. Among these, sustainability considerations, such as the integration of renewable energy sources, are gaining importance, yet they introduce new challenges traditionally absent from energy minimization approaches. Indeed, the growing integration of renewable energy sources brings variability and uncertainty that must be managed alongside service-level requirements such as stability and robustness. This variability complicates orchestration decisions, especially in dynamic and heterogeneous environments where both voluntary and involuntary node departures may occur. Orchestration represents both a challenge and a lever for the energy management of future systems of this type that will be deployed on a global scale. Faced with current environmental concerns and the growing and plethora of heterogeneous and scattered resources involved in these systems, it is imperative to rethink the design of orchestration methods from the additional perspective of sustainability and efficiency.

In this thesis, we present an in-depth investigation of service orchestration in renewable energy-enabled edge computing environments. First, we introduce a comprehensive classification framework specifically designed for sustainable Edge service orchestration across traditional and renewable energy contexts. This review strengthens the critical importance of addressing sustainability concerns and demonstrates how virtualization and service orchestration constitute fundamental foundations for future energy-efficient, distributed ICT infrastructures. Then, we propose a system model that captures the interactions among renewable energy generation, Edge infrastructures, and service robustness. Building on this model, we design an algorithm for service placement and placement reconfiguration, using predictions of energy availability and workload demand to enable more efficient and sustainable orchestration decisions. Finally, we introduce an architecture that instantiates the proposed theoretical models into a practical, standards-compliant solution. Collectively, these contributions provide novel and effective approaches to the challenge of service orchestration in renewable energy-enabled Edge computing settings.

Keywords : Edge computing, Service orchestration, Sustainability, Energy efficiency, Renewable energy

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Contexte au sein du PEPR Réseaux du futur

Dans le cadre de son programme de recherche scientifique France 2030, le gouvernement français a lancé plusieurs programmes PEPR (Programmes et Equipements Priotaires de Recherche) thématiques dans différents domaines scientifiques. L'un d'entre eux est consacré à la 5G et aux réseaux du futur (NF). Au sein de ce programme, NF-HiSec vise à traiter les questions de sécurité des réseaux du futurn, en particulier la 5G. Le projet NF-HiSec cherche à relever les principaux défis en matière de sécurité en se concentrant sur cinq objectifs principaux :

— **Détection et atténuation des attaques dans des environnements complexes :** Développer des techniques avancées pour identifier et atténuer les cyberattaques dans des réseaux dynamiques et en constante évolution.

— **Protection des données personnelles, avec un accent sur l'interception légale :** Sécuriser les données personnelles même lors d'interceptions légales, tout en garantissant le respect des lois sur la vie privée.

— **Modélisation des mécanismes de sécurité pour répondre aux besoins des applications :** Adapter les mesures de sécurité aux exigences spécifiques des différentes applications utilisant ces réseaux.

— **Intégration de la sécurité à travers les frontières matériel et logiciel :** Formaliser la connexion entre les composants matériels et logiciels afin d'assurer la mise en œuvre de mesures de sécurité complètes à tous les niveaux.

Dans le cadre de cette large initiative, le projet NF-HiSec joue donc un rôle central dans le développement de méthodes et d'outils innovants visant à sécuriser de manière exhaustive les réseaux du futur. Cette thèse s'inscrit ainsi comme une contribution à un effort commun, cherchant à sécuriser une partie de l'ecosystème divers et complet qu'est la 5G.

Plan

Chapitre 1

Définitions des surfaces d'attaques de la 5G

Sommaire

Les réseaux 5G sont conçus pour répondre à des exigences plus strictes en termes de débit amélioré (eMBB), de latence ultra faible (URLLC) et de communication massive d'appareils connectés (mMTC), permettant l'émergence de nombreux services innovants[**?**, **?**]. Pour assurer la mise en œuvre et le fonctionnement efficace de ces différents cas d'usage, la 5G change profondément de paradigme et intègre plusieurs nouveaux concepts aux réseaux mobiles, de nouveaux composants dans son architecture et de nouvelles méthodes de communications entre les entitées qui la compose.

Ces évolutons rendent les réseaux mobiles plus flexibles et plus dynamiques mais au prix d'une complexification du système et d'une augmentation de la surface d'attaque. Dans ce contexte, les besoins en matière de détection d'anomalies évoluent également et il devient alors nécessaire de concevoir de nouveaux mécanismes de détection capables de s'adapter aux spécificités et à la complexité des architectures 5G.

1.1 Changements de la 5G en regard de versions précédentes

1.1.1 Virtualisation des fonctions réseaux et changement de paradigme

La virtualisation constitue la base technologique qui permet des fonctions avancées telles que le network slicing. Elle repose sur deux concepts clés, NFV et SDN, qui permettent de déployer des services à la demande sur tout type de matériel. Grâce à ces approches, les réseaux 5G s'éloignent de la conception traditionnelle des systèmes mobiles, historiquement construits sous forme d'architectures monolithiques où chaque fonction s'exécutait sur un matériel dédié et communiquait via des canaux fixes et des piles de protocoles rigides. À la place, l'architecture 5G adopte une approche plus proche des microservices du cloud, où les fonctions réseau sont softwarisées, découplées et conteneurisées.

Cependant, malgré les avantages significatifs de la virtualisation, celle-ci hérite également d'un nombre important de nouvelles vulnérabilités propres à cette technologie. En effet, qu'elle repose sur des technologies de machines virtuelles ou sur la conteneurisation, la virtualisation représente une couche supplémentaire qui peut elle-même être vulnérable à des attaques spécifiques tel que l'empoisonnement d'image [?] ou l'évasion de VM/Docker [?, ?].

1.1.2 Evolution des interfaces et piles protocolaires

Certains protocoles et interfaces hérités de la 4G sont conservés en 5G, ce qui implique que cette dernière hérite de la surface d'attaque et de certaines vulnérabilités des générations précédentes. On retrouve par exemple des attaques d'injection de messages de contrôle dans le flux utilisateur géré par le protocole GTP-U [?, ?], des attaques par inondation ciblant la gestion des ressources radio via le protocole RRC [?], ainsi que des attaques par rejeu affectant les messages de signalisation du protocole NAS [?].

Mais la transition vers une architecture inspirée des microservices introduit aussi de nouvelles interfaces et de nouveaux protocoles, élargissant donc par conséquent la surface d'attaque. L'adoption des Service-Based Interfaces, basées sur des API REST, expose notamment le cœur de réseau à des attaques de type fuzzing [?]. De nouveaux protocoles comme NGAP peuvent être ciblés par des injections de messages malformés [?], tandis que PFCP introduit de nouvelles surfaces d'attaque liées à la gestion des sessions et des flux [?].

1.1.3 Développement des réseaux privés

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1.1.4 Ecosystème élargi et nouveaux vecteurs d'intrusions

La softwarisation de la 5G et l'introduction de nouveaux mécanismes aux réseaux mobiles ajoute de nouvelles problématiques de sécurité et introduit de nouveaux vecteurs d'intrusion. Parmi les nombreuses nouveautés que la 5G apporte, voici quelques exemples notables qui contribuent à l'élargissement de la surface d'attaque.

— **Slicing** : Afin de gérer efficacement plusieurs services et cas d'usage, la 5G introduit le concept de network slicing qui consiste à créer plusieurs réseaux virtuels au sein d'une même infrastructure physique partagée. Le concept de slicing impacte l'architecture 5G à tous les niveaux, depuis la couche physique, où il implique la gestion de la grille de ressources temps/fréquence [**?**], jusqu'à la couche applicative, où il garantit la continuité et l'isolation des services. Bien que les spécifications du 3GPP relatives aux slices intègrent des mécanismes de sécurité robustes [**?**, **?**], les implémentations actuelles, en particulier open source, restent encore immatures. Cette situation soulève des préoccupations de sécurité et introduit de nouveaux vecteurs d'attaque, notamment liés à une isolation imparfaite et au partage de ressources physique.

— **Orchestrations** : Dans un déploiement à grande échelle, il est n'est pas réaliste de gérer l'entiereté de l'infrastructure des réseaux 5G manuellement. La virutalisation des fonctions réseaux permet cependant d'automatiser l'orchestration et la gestion des services [**?**] pour pouvoir moduler, en fonction des besoins, l'architecture et les capacités du réseaux. Cette gestion automatisée est généralement réalisée par des logiciels comme Kubernetes ou Openstack qui rajoutent des couches de complexité et de vulnérabiltiés.

— **Chaîne d'approvisionnement** : L'intégration logicielle était historiquement gérée par un acteur unique : le fournisseur d'équipements réseau. Cependant, les réseaux 5G reposent maintenant sur un bien plus grand panel de logiciels pour leur fonctionnement. Comme c'est généralement le cas dans le développement logiciel moderne, ces logiciels reposent largement sur du code tiers, dont une grande partie est open source. Cela introduit des risques liés à la méconnaissances de l'inventaire détaillé des composants logicielle (Software Bill of Materials) ainsi qu'aux attaques de la chaîne d'approvisionnement [**?**, **?**].

1.2 Infrastructure et Composants Physiques

L'architecture 5G repose sur un ensemble distribué de composants interconnectés, organisés en plusieurs sous-systèmes coopérant pour assurer la connectivité et la gestion des communications. Bien que de nombreux autres éléments existent, les composants présentés ci-dessous comptent parmi ceux absoluments essentiels au fonctionnement du réseau 5G. La Figure **??** en propose une illustration simplifiée.

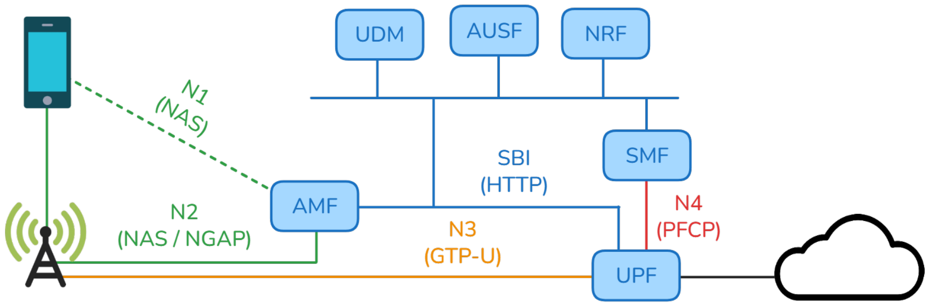

FIGURE 1.1 – Illustration simplifiée des principaux composants de l'architecture 5G

1.2.1 Equipements Utilisateurs (UE)

Les UE désignent les dispositifs qui se connectent au réseau 5G, tels que les smartphones, les objets connectés et, plus généralement, tout équipement disposant d'une interface radio par laquelle ils peuvent communiquer avec le réseaux 5G. Ils constituent la grande majorité des entités actives du réseau et représentent, par nature, les composants les plus exposés de l'infrastructure. En raison de leur nombre, de leur vulnérabilité et de leur accessibilité, ils sont à la fois les cibles mais aussi les vecteurs les plus probables lors d'une attaque [**?**, **?**, **?**].

Du point de vue d'un attaquant, compromettre un UE peut s'avérer relativement simple. L'accès peut être réalisé à distance, par installation de logiciels malveillants, ou physiquement, en dérobant l'appareil la carte SIM qui l'authentifie. Bien que les UE ne soient capable d'intéragir directement qu'avec la station de base, ils peuvent par exemple être exploités pour mener des attaques par déni de service distribué (DDoS).

1.2.2 Réseaux d'accès radio (RAN)

Les UE se connectent au cœur de réseau via le RAN, composé d'un ensemble de stations de base radio, appelé gNB et éventuellement complété par d'autres relais radio. Le gNB assure l'interface avec le cœur de réseau via l'AMF, ainsi qu'avec le réseau de données via l'UPF auxuquel il est connecté par liaison IP standard. Le gNB prend en charge les communications radio, la gestion de l'accès et de l'ordonnancement des ressources, et garantit la continuité de service grâce aux mécanismes de handover entre stations de base voisines.

Contrairement aux réseaux 4G, les réseaux 5G reposent sur un déploiement beaucoup plus dense de stations radio, situées plus près des utilisateurs finaux. Cette proximité accrue facilite l'accès physique pour des attaquants susceptibles de tenter de prendre le contrôle de ces équipements. De plus, les gNB sont relativement faciles à usurper pour un adversaire en raison de l'absence de mécanisme d'authentification forte vis-à-vis des UE. En effet, en émettant un signal plus puissant que celui des stations de base légitimes à proximité, un attaquant peut inciter les terminaux environnants à se connecter à leur station malhonnête en priorité[?]. Ces attaques sont d'autant plus facilitées par la disponibilité croissante d'outils de radio définie par logiciel (SDR), qui nécessitent relativement peu de ressources et de compétences techniques pour être utilisés.

1.2.3 Coeur de réseaux (CN)

Le cœur de réseau (CN) est un ensemble de fonctions réseau interconnectées via une infrastructure IP, qui interagissent entre elles ainsi qu'avec les UEs et les gNB à travers des chaînes de procédures. Ces fonctions assurent chacune un rôle spécifique et un grand nombre d'entre elles sont définies par le standard 3GPP. Certaines sont essentielles au fonctionnement du réseau, tandis que d'autres répondent à des cas d'usage plus spécifiques et ne sont donc pas systématiquement déployées. Parmi les fonctions réseau essentielles, on retrouve notamment :

- **AMF (Access and Mobility Management Function)** : gère l'enregistrement des UEs, la mobilité et la signalisation NAS.
- **SMF (Session Management Function)** : responsable de l'établissement, de la modification et de la libération des sessions PDU.
- **UPF (User Plane Function)** : assure le transfert des données utilisateur et le routage vers les réseaux externes.
- **AUSF (Authentication Server Function)** : réalise les procédures d'authentification des utilisateurs.
- **UDM (Unified Data Management)** : gère les données d'abonnement et les profils des utilisateurs.
- **NRF (Network Repository Function)** : permet la découverte et l'enregistrement des fonctions réseau.

Le CN est significativement plus difficile d'accès que les UEs ou les gNBs, car les machines physiques qui les hébergent ne sont généralement pas exposés à un accès physique direct. Néanmoins, en raison de leur forte valeur stratégique, compromettre une fonction réseau reste un objectif réaliste pour un adversaire déterminé. Comme évoqué dans les sections précédentes concernant l'orchestration et la chaîne d'approvisionnement, il demeure possible d'atteindre le cœur de réseau pour compromettre certains de ses éléments et obtenir un accès direct au reste du CN. Cet accès pourrait notamment être utilisé pour altérer l'architecture du cœur de réseau en ciblant le NRF[?], exposer des données utilisateur via l'UDM[?], et perturber la gestion des sessions en interférant avec la communication entre le UPF et le SMF[?].

1.3 Définition du cadre d'étude de la thèse

Comme présenté dans les sections précédentes, l'écosystème et la surface d'attaque de la 5G se sont considérablement élargis par rapport aux générations précédentes, formant plusieurs domaines hétérogènes et attaquables à différents niveaux. Ces domaines sont vastes et présentent peu de recouvrement. Il n'est donc pas envisageable de sécuriser l'ensemble de la 5G et il est plutôt nécessaire de se focaliser sur des problématiques ciblant des parties restreintes de cet écosystème. La virtualisation et la softwarisation constituent l'un des changements majeurs de cette génération, qui se reflètent principalement sur les couches protocolaires hautes. C'est pourquoi nous avons choisi de concentrer le travail de cette tèse sur la partie applicative du trafic de contrôle transitant par le cœur et le RAN de la 5G.

Les sections précédentes ont montré que la 5G est conçue pour être très modulaire, avec des NF pouvant être définies de manière arbitraire selon les besoins, notamment dans le cas des déploiements privés. Nous restreignons donc notre champ d'étude à un réseau comprenant uniquement les NF essentielles et géré par un seul opérateur. Nous excluons aussi les problématiques liées au network slicing, à la virtualisation ou à l'orchestration, qui nécessiteraient une analyse spécifique. Nous considérons plutôt ces mécanismes comme des vecteurs potentiels d'intrusion qui pourrait permettre à un attaquant de compromettre un ou plusieurs composants du réseau.

Comme nous n'avons pas d'informations sur l'état de maturité ni sur les problématiques de déploiement rencontrées par les opérateurs publics, notre étude se base à la fois sur une version idéale de la 5G telle que définie par la norme 3GPP et sur les implémentations open source disponibles publiquement mais qui ne suivent pas encore fidélement les standards. Nous supposons que ces implémentations open source sont bien moins matures que les déploiements réels des opérateurs commerciaux, mais nous faisons l'hypothèse que, étant donné la complexité du système, même les réseaux 5G des opérateurs doivent s'écarter du standard à certains égards. De plus, dans le cadre de déploiements privés, il est tout à fait plausible que de petits groupes utilisent directement ces plateformes open source.

Chapitre 2

Classification of service orchestration strategies

This chapter addresses the problem of energy efficiency in the orchestration of Edge services. Orchestration represents both a challenge and a lever for the energy management of future systems of this type that will be deployed on a global scale. Faced with current environmental concerns and the growing and plethora of heterogeneous and scattered resources involved in these systems, it is imperative to review the design of orchestration methods from the additional perspective of sustainability and efficiency. We provide a comprehensive classification framework specifically designed for energy-efficient Edge service orchestration, across traditional and renewable energy contexts. Our analysis reveals three distinct categories of traditional strategies and identifies three emerging orchestration paradigms unique to renewable energy integration. We explore the research landscape, identifying strategies that consider renewable energy, energy storage, and renewable energy management. We also provide three case studies and discuss eleven potential future directions.

Sommaire

2.1 Energy-efficient service orchestration strategies

Service orchestration at the network's Edge needs to find balance between optimizing energy consumption and ensuring operational performance. In this section, we address traditional energy consumption reduction strategies, categorized in three parts : resource management, hardware management, and service management.

2.1.1 Resource management

Resource management plays a critical role in optimizing the performance and energy-efficiency of distributed systems. In the context of Edge computing, resource management refers to the process of efficiently allocating, monitoring, and optimizing computational and network resources across distributed Edge devices. Given the increasing demand for low-power solutions, it is crucial to consider the management of Edge resources within the scope of sustainable service orchestration. In this section, we classify resource management strategies into three categories : computation, communications, and caching/data storage.

2.1.1.1 Computation

With Cloud-Edge technologies, each deployed service has an associated processing cost. This cost is typically linked to the execution time required for a given task, which depends on the hardware, usually the CPU (or GPU) used to process it. Indeed, these components have specific processing capacities that determine the speed at which a workload can be handled. Most computational models use CPU frequency, which is the number of operations the CPU can execute in a set time interval, as seen in works such as [?, ?, ?, ?, ?, ?, ?, ?]. A frequent model for computing the CPU energy cost of a task is given using (??), with κ being the effective switching capacitance based on the CPU's architecture, f the CPU's current frequency, D the amount of operations to complete, and finally T the total time to process the workload.

$$EC_{computing} = \kappa f^2 D = \kappa f^3 T. \tag{2.1}$$

Other models in the literature may directly use a factor representing the energy consumed per unit of operations, as highlighted in works like [?, ?, ?] or the time spent in active state [?]. Models considering an estimated usage per service can also be used, as in [?]. Computation on Edge nodes often involves a mix of static energy costs (baseline energy needed to keep devices operational), and dynamic energy costs (additional energy consumed while processing tasks).

Some works rely on real-time consumption measurements, such as those presented in [?, ?]. While the use of real-time measurements could seem advantageous compared to models, these measures could interfere with other consumption metrics, leading to potential inconsistencies in data interpretation.

With energy consumption being linked to the use of computational resources, reducing or optimizing their usage is an effective way to minimize it.

2.1.1.2 Communications

The second key resource management strategy in Edge computing is communications. It complements computing by representing the network cost. Most models in the literature take into account both transmission power and transmission delay (**??**) when assessing communication costs, as seen in works like [**?**, **?**, **?**, **?**]. The transmission power p refers to the electrical power (usually in watts/milliwatts), that a transmitter uses to send a signal through a communication channel. Transmission delay T refers to the time it takes for the device to physically transmit the data, usually expressed as amount of data D by maximum data rate R. The latter is typically calculated using Shannon's formula (**??**), as demonstrated in [**?**, **?**, **?**, **?**, **?**]. This formula, which includes the signal-to-noise ration (SNR), defines the capacity of a communication channel by comparing the desired signal level to the background noise.

$$EC_{communications} = p \times T = p \times \frac{D}{R} \tag{2.2}$$

$$R = B \times log_2(1 + SNR) \tag{2.3}$$

In addition to transmission power, the energy cost of data transmission is frequently tied to the volume of data transferred. This approach is employed in studies such as [**?**, **?**]. In some models, researchers also account for static circuit power consumption, as seen in [**?**, **?**]. Some works also consider the physical distance between nodes, as [**?**].

In summary, network energy consumption primarily results from large data volumes or limited bandwidth, making it crucial in the Cloud-Edge continuum to make aware decisions about when and where data transmission is necessary.

2.1.1.3 Caching & Data storage

Caching is a popular technique used in the Cloud-Edge continuum to improve performance, reduce latency, and optimize resource usage. It involves temporarily storing frequently accessed or recently used data in a high-speed storage layer called a cache. This cache allows new requests for the same data to be served faster, as retrieving it from the cache is quicker than fetching it from the original source. The energy cost of caching is studied in [**?**, **?**]. In those cases, each Edge node has a maximum caching capability. In [**?**], the cost of storing data is evaluated by the energy consumption of SSD read/write operations, which is determined by the device's read and write speeds.

Caching is particularly valuable in scenarios where the same data is requested repeatedly, or where the cost of recomputing or retrieving data is high. However, caching also introduces challenges such as ensuring data consistency (cache coherence) and managing cache size and eviction policies.

2.1.2 Hardware management

Effective hardware management is essential for energy-efficient service orchestration in Edge computing environments. Given the limited resources and power constraints of Edge devices, optimizing hardware usage plays a crucial role in reducing overall energy consumption while maintaining service performance. In this section, we explore three key hardware management techniques that significantly impact energy efficiency : sleep mode management, DVFS and device-related energy consuming activities.

2.1.2.1 Sleep mode management

Sleep-awake mechanisms can dynamically switch between sleep and active states based on current load. Sleep mode refers to low-power state where the device reduces its active components, thereby consuming minimal energy while remaining ready to resume operations quickly when needed. This is particularly important in distributed systems because of the number of employed devices.

For instance, [?] introduces a MEC network where base stations can enter sleep mode to save energy. However, this energy saving must be balanced against the costs of frequent state switching, which can cause network instability and hardware degradation. The authors model this trade-off through a base station activation cost.

2.1.2.2 Dynamic Voltage & Frequency Scaling

DVFS is a widely adopted technique for optimizing energy consumption in computing systems, particularly in Cloud environments. DVFS allows the operating voltage and frequency of the CPU to be adjusted dynamically based on the current workload, resulting in significant power savings without sacrificing performance. By scaling down the frequency and voltage when full computational power is not required, systems can reduce their energy consumption while still meeting task demands. In [?], the authors employ a double deep Q-learning model to choose which DVFS algorithm to use. In [?], DVFS is used in combination with a task scheduling mechanism.

2.1.2.3 Device-related energy consuming activities

In addition to the energy consumed directly by services in Edge computing, several auxiliary processes can also contribute to the overall device energy footprint.

These processes, although not directly tied to service execution, must be considered into energy-efficient service orchestration strategies to optimize total system performance.

For example, in scenarios involving mobile nodes, energy consumption extends beyond the computational tasks. In [?], UAV's propulsion energy consumption is incorporated into the overall energy optimization strategy. Similarly, [?] considers UAV activity by modeling hovering power consumption in their energy minimization problem.

Another example of non-service-related energy consumption can be found a smart farming application. In [?], energy consumption is not only linked to IoT data processing but also to the use of high-voltage pulses for insecticidal activities. Additionally, the system makes use of weather forecasting to determine the most energy-efficient timing for pest control operations. Similarly, in [?], the brightness of solar-powered street lamps can be adjusted to minimize the energy consumption. These processes, while not related to direct service execution, are essential to consider in an energy-efficient orchestration of Edge applications.

2.1.3 Service management

Service management can be considered as all the techniques that aim at optimizing the deployment, execution, and maintenance of computational tasks, applications or functions across the network. These techniques include service placement, task scheduling, task offloading, queue management, and service migration. It's important to note that these approaches often overlap and are frequently used in combination to address the complex dynamic nature of Edge environments.

2.1.3.1 Service placement

Service placement aims to optimize the geographical or relative positioning of services across distributed Edge nodes. Effective placement decisions are often tied with resource allocation strategies to ensure that services meet performance requirements, while also minimizing energy consumption. The strategic deployment of services helps balance loads and costs, especially in scenarios where scenarios need to communicate across the network.

Communication between service components is addressed in [?], where the authors consider functions-as-a-service (FaaS) with applications being constituted of multiple functions in a chain that can be placed separately. Similarly, [?] optimizes service placement across Edge servers by considering multiple features including geographical coordinates, user distribution patterns, and service preferences.

2.1.3.2 Task scheduling

Task scheduling in Edge computing refers to the process of allocating and timing operations on Edge resources in a way that meets specific requirements (e.g. deadlines).

In task scheduling, minimizing energy consumption while ensuring timely task execution is a critical challenge. For instance, [?] proposes a task scheduling algorithm that reduces energy consumption and latency, and increases as well the system reliability. Similarly, the authors of [?] address energy-efficient scheduling by using a DRL-based algorithm that learns an optimal strategy to balance energy consumption and execution time. In [?], the authors address a job scheduling problem with three components : job mapping, resource allocation, and operation timing determination for offloading, processing, and downloading phases. The optimization minimizes energy consumption while satisfying machine capacity, resource availability and timing constraints.

Scheduling can also be applied to services, and it differs from service placement by coordinating the execution of services over a defined period, instead of finding the optimal location for the deployed services.

2.1.3.3 Task offloading

Task offloading in Edge computing is closely related to task scheduling but is more distributed in nature. In offloading, users can decide whether to process tasks locally on their devices or offload the computation to an external system, such as a nearby Edge server or Cloud infrastructure. This decision depends on factors like resource availability, energy consumption, and processing power.

In [?], a method for asynchronous mobile edge computation is introduced, where tasks can be processed locally or offloaded to an Edge Cloud. The authors of [?] propose a centralized scheduling approach that allows end devices to submit composite tasks, which are then intelligently dispatched across Edge servers and Cloud. Finally, in [?], the concept of offloading tasks from vehicles to other vehicles, Edge servers, or to the Cloud via Edge devices is explored.

2.1.3.4 Queue management

Queue management can be used in Edge computing systems, particularly in task scheduling and congestion control scenarios. It helps in organizing and prioritizing tasks, managing system resources efficiently, and minimizing delays. The study presented in [?] employs an M/G/1/PR queuing model to represent workload distribution. This model is integrated with a deep deterministic policy gradient (DDPG) algorithm to schedule tasks over distributed data centers. Effective queue management in Edge computing environments faces several challenges. These include the heterogeneity of tasks and resources, the dynamic nature of task arrivals and service times, and the need to balance multiple objectives such as minimizing delay, maximizing throughput, and optimizing energy efficiency.

2.1.3.5 Service migration

Service migration involves the dynamic relocation of any kind of service (computational tasks, applications, VMs, etc.) across different nodes in the network. This process is a key lever for maintaining performance, optimizing resource utilization, and improving energy efficiency.

The necessity for service migration often comes from various factors, including user mobility, changing network conditions, and dynamic resource availability. In [?], digital twins are migrated when the association changes incurring a migration cost. Energy efficiency is a primary driver for service migration in many scenarios. In [?], the researchers consider VM migration in the context of user mobility, aiming to minimize energy consumption as users move across the network. Similarly, the authors of [?] propose a multiple service migration optimization problem, notably reducing the energy consumption of the process.

Service migration energy costs are typically based on data migration, which involves transferring the state and data of the service from one node to another, and service startup, which includes the time and resources required to initialize the service at its new location. These costs must be carefully weighed against the potential benefits of migration to ensure that the migration decision enhances overall system performance.

2.2 Novel service orchestration strategies

Incorporating renewable energy into service orchestration for Edge computing introduces a shift from the traditional balance between energy consumption and performance to a more complex optimization challenge. These new strategies do not replace the traditional ones, but instead add a new layer of complexity to them. As a result, optimization must now explicitly account for energy variability. Unlike conventional systems, where the goal is often to minimize energy usage, the integration of renewable energy requires a focus on efficiently managing a limited and variable energy supply over time. Renewable energy sources, like solar or wind, are inherently unpredictable and impacts system stability and reliability. This variability necessitates new strategies to ensure consistent performance while making the most of the available energy. Moreover, service orchestration in this context evolves the traditional multi-objective optimization problem, introducing additional objectives and constraints.

In this section, we will explore the novel approaches introduced by integrating renewable energy sources into the orchestration process to understand how renewable energy integration changes orchestration. We introduce again three categories, namely energy harvesting, renewable energy storage, and renewable energy management.

2.2.1 Energy harvesting

Energy harvesting refers to the process of capturing and converting ambient energy from the environment, such as solar, wind, or radio frequency (RF) signals, into electrical energy. In Edge computing, it can be used to power remote Edge nodes, that might operate in environments where batteries or wired power are infeasible.

2.2.1.1 Renewable energy sources

Solar energy is one of the most widely adopted renewable energy sources in Edge computing and IoT environments due to its availability and the decreasing costs of photovoltaic systems. In the Edge layer, solar power allows Edge nodes, which are often distributed in remote locations, to operate independently of traditional power grids, which are usually powered totally or partially by fossil energy. This decentralization is crucial for maintaining continuous operations in areas where energy infrastructure is limited or unreliable, as well as reducing reliance on fossil fuels, decreasing the carbon footprint of Edge operations. Studies [?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?] make use of solar power in their architecture. Moving up to IoT/terminal layer, solar energy is primarily used to power sensors like in [?] and IoT-based systems as in [?, ?], allowing continuous operation in resource-constrained environments. These devices benefit from solar power's reliability, particularly in remote deployments. It can be extended to user equipment as in [?]. Solar-powered charging stations in [?] further expand the potential of solar energy, offering sustainable charging options for electric vehicles, while in mobile Edge networks, operators benefit directly from solar-driven vehicles for sustainable operations [?].

Wind energy serves as a complementary renewable energy source, particularly when combined with other intermittent renewable sources, such as photovoltaic panels. Like solar, wind energy harvesting is widely applied across different levels of Edge and IoT infrastructures. In the Edge layer, wind energy facilitates autonomous operations and energy management systems like in [?, ?, ?, ?, ?, ?, ?]. Wind energy also powers distributed user-centric systems, where it helps with the management of energy demands for mobile users [?]. For vehicular infrastructures, wind power is critical in operating sustainable charging stations for mobile Edge devices as in [?] and [?].

Solar and wind energy are frequently deployed complementing each other to address their inherent intermittency. Solar power generation peaks during daylight hours, particularly around midday, but drops off significantly at night or during cloudy conditions. On another hand, wind energy production is highly variable, and might be peaking at night or during stormy weather when solar output is minimal. By combining both technologies, energy systems can achieve a more stable and reliable power supply.

Several other forms of renewable energy are being explored, but often restricted to the IoT/terminal layer due to their low power generation :

- *Piezoelectric energy :* Energy is harvested from mechanical vibrations or movements, as mentioned in distributed offloading solutions in [?].
- *Kinetic tiles :* Kinetic energy harvesting from human motion, such as in public spaces, is another emerging renewable source cited in [?].

2.2.1.2 Radio frequency signal harvesting

RF signal harvesting is a form of energy harvesting that involves capturing ambient RF energy from wireless signals, such as Wi-Fi or mobile communication signals. It is there again used mainly to power small devices. This strategy can be divided in two main categories :

- *Ambient RF Signals :* This technique is useful for IoT and sensor devices, which can harvest RF energy to remain operational in low-power environments like in [?, ?].
- *Wireless Power Transfer :* Wireless power transfer (WPT), or sometimes wireless energy transfer (WET), allows energy to be transmitted wirelessly to Edge or IoT devices, ensuring continuous operation as in [?, ?, ?, ?].

Energy harvesting techniques, especially those other than solar panels and wind turbines, are primarily designed for low-power devices, making them more appropriate for the IoT/terminal layer. However, in Edge environments, the deployment of larger renewable sources like solar and wind remains viable. Other potential sources, such as hydroelectric power, might be employed, but they are constrained by their complex setup and geographic dependencies.

In early works, energy harvesting is largely treated as a means to offset the energy consumption of Edge nodes, as in [?, ?]. More recent studies, however, demonstrate that renewable energy availability can also serve as a driving factor for service migration decisions, as shown in [?, ?].

2.2.1.3 Renewable energy implementations

In the literature, renewable energy sources, especially solar and wind, are integrated through various modeling techniques to account for their unpredictable nature of due to changing environmental conditions. To model this behavior, stochastic processes are a popular practice. A common approach is to simulate energy availability using uniform distributions, which assume that energy generation fluctuates evenly over time between two boundaries. This approach is seen in studies [?, ?, ?, ?, ?]. Other works model intermittency using a Bernoulli distribution, which represents energy availability as a series of binary outcomes, as seen in [?]. Additionally, standardized Gaussian and normal distributions are employed to introduce variability in energy generation around a mean value, as explored in [?, ?, ?]. These models help to simulate and optimize the use of renewable energy in Edge systems by addressing the stochastic nature of energy production. The work in [?] focuses on space-based networks, where the authors specifically examine seasonal impacts on sunlit ratios.

Another approach to incorporate renewable energy in Edge service orchestration involves using real-world energy generation traces data from solar and wind sources. These data captures historical power measurements from actual solar panels or wind turbines, providing realistic input for simulations or emulations of Edge networks. For solar energy, studies [?, ?, ?, ?, ?, ?, ?] have used solar trace data in their work. Similarly, wind energy traces are employed to assess Edge system performance under varying wind conditions, as shown in works like [?, ?]. By using these traces, researchers can better predict energy availability, leading to more efficient resource management in Edge networks powered by renewable energy.

Finally, some studies have moved beyond simulations to implement renewable energy directly into Edge environments. For instance, small solar cells with maximum output of 3W have been deployed in real-world scenarios to power IoT devices in the energy-efficient Edge systems of [?]. A larger scale implementation, the Cellnex Mobility Lab in Castellolí (Spain), described in [?], combines solar panels and wind turbines to power an Edge computing infrastructure, demonstrating the potential of renewable energy to sustain Edge services in real-world conditions. These implementations serve as proof-of-concept for how solar and wind energy can be integrated into Edge networks, offering insights into practical challenges and solutions for renewable energy deployment.

2.2.2 Renewable energy storage

Energy storage is often associated with the use of rechargeable batteries. Batteries play a central role in storing electrical energy with a finite capacity and releasing it to power various devices and systems. Batteries are not the only method for energy storage though, as other approaches such as water-based storage (e.g., pumped-storage hydroelectricity as mentioned in [?]) or kinetic energy storage (e.g., flywheels) exist but are beyond the scope of our work.

2.2.2.1 Advantages of batteries with renewable sources

Batteries offer several key advantages when integrated with renewable energy sources, addressing some of the primary challenges associated with intermittent energy production. One of the most significant benefits is their ability to mitigate the intermittency of renewable energy generation, as they might depend on environmental conditions for example. They improve flexibility by allowing energy to be stored when production exceeds demand and then consumed during peak periods. This ability to balance load across time helps to avoid overloading the power infrastructure and makes energy distribution more efficient. Additionally, batteries play a crucial role in enabling Edge nodes, especially those deployed in remote or mobile environments, to operate independently without needing constant connection to the power grid. This autonomy is essential in regions where infrastructure is limited or unstable, as well as in mobile applications where nodes may need to move frequently and remain disconnected from traditional energy sources. Similarly, batteries also serve as a reliable emergency backup system. In cases where the primary power supply is disrupted or unavailable, stored energy can be released to maintain the functioning of critical systems. This capability ensures continuous operation during outages and enhances the resilience of the overall system as demonstrated in [?]. To sum things up, the use of batteries promotes energy independence, reducing reliance on external energy sources and assuring a more self-sustaining, renewable-powered infrastructure.

2.2.2.2 Battery modeling

The most common battery model we extracted from the literature can be generalized to (??). $B(t)$ represents the battery level at time t, $EC(t, t+1)$ refers to the energy consumption over the interval $[t, t+1]$, and $EH(t, t+1)$ to the energy harvested during the same interval. The battery's capacity being finite, it is bounded by the relationship $0 \leq B(t) \leq B_{max}$ where B_{max} is the maximal capacity of the battery. In some works [?, ?, ?], this model is mentioned as "battery energy queue". In Edge computing scenarios, if the battery is fully discharged, the node may no longer be supplied with energy unless another energy source is available. If the battery becomes fully charged and cannot store additional energy, excess energy may be wasted unless it is diverted to other locations or purposes as in [?].

$$B(t+1) = B(t) - EC(t, t+1) + EH(t, t+1) \tag{2.4}$$

Additionally, service orchestration in Edge computing frequently incorporates battery status into decision-making processes. For instance, in [?], task allocation and orchestration are managed while considering the current battery levels to optimize energy usage and ensure longevity. Similarly, in studies like [?, ?, ?], orchestration strategies are "battery-aware", meaning that their battery state of charge is incorporated into their decision process. Also, battery stability is treated as an optimization goal in various studies. In [?, ?], ensuring that the battery remains within an optimal charge range is a key objective to prevent disruptions and extend service availability.

Though simple models dominate the orchestration literature, some studies might propose more advanced battery models. For example, [?] introduces a discount factor to account for natural energy decay in batteries. Similarly, in [?], the energy depletion rate and the battery health index are used. While most cited works rely on the simplified battery model of equation (??) for service orchestration, this approach fails to capture the complete reality of battery behavior, as real batteries experience natural energy loss and capacity degradation over time.

2.2.2.3 Battery management

Battery management is a critical aspect of energy storage in Edge computing, focusing on optimizing the use of batteries to make a more efficient use of energy and increase the longevity of the energy storage system. In the literature, various strategies are employed to manage battery resources effectively, ranging from lifespan prediction to energy storage optimization.

A key aspect of battery management is predicting battery lifespan, which helps to take orchestration decisions and ensures more efficient energy use. For instance, in [?], a Lenovo Smart Power Agent is used to forecast the remaining battery life, allowing the local planner to make pertinent decisions about task scheduling and resource allocation. Similarly, [?] extends the battery lifetime by considering the sunlight and being able to minimize the use of each satellite battery. [?] estimates battery lifespan by analyzing energy consumption across multiple scenarios, enabling better planning for energy use. Additionally, [?] focuses on estimating the charging duration for solar-powered insecticidal lamp batteries, to optimize pest eradication in intelligent farms. [?] considers operational and maintenance costs, as well as a battery aging cost.

Another critical factor is battery protection. In [?], techniques are mentioned to prevent damage to batteries, such as safeguarding against overcharging, overload and overvoltage, as well as discharging strategies to extend battery life by preventing burst depletion. These protective measures are essential for maintaining the health of the battery and ensuring its long-term performance in Edge environments, as a decreased battery's health lead to a decreased capacity. The authors in [?] designed a battery health monitor that optimize the discharge rate to maximize the battery life by tracking the battery status, energy depletion rates, temperature fluctuations and usage intensity. In [?], the authors focus on minimizing availability variance among nodes to reduce node failure rates and prevent the waste of renewable energy inputs. By ensuring that energy distribution across nodes is balanced, this strategy helps maximize the efficiency of the overall system and minimizes energy loss due to unused resources.

Battery management mainly involves making decisions about when and how much energy to store. For instance, in [?], energy storage strategies are explored to ensure that the battery maintains sufficient charge levels and store energy during periods when it is cheaper or more abundant. Similarly, [?] investigates how to determine the optimal amount of energy to store, balancing the need for immediate use with future energy demands.

These various approaches to battery management are crucial in ensuring that Edge computing systems remain resilient, energy-efficient and sustainable, especially when relying on renewable energy sources.

2.2.2.4 Specificity of vehicular networks

Vehicular networks, which generally include electric vehicles (EVs) and unmanned aerial vehicles (UAVs), are a well investigated subject in modern Edge computing systems. These networks consist of mobile nodes that either serve as or rely on Edge infrastructure to process and store data close to where it is generated. In Edge computing, proximity to the data source is crucial, so vehicular networks provide a dynamic and distributed approach to this need, especially in areas where traditional infrastructure is lacking or insufficient.

UAVs and EVs, equipped with batteries, can support a range of applications in

Edge computing. UAVs, for instance, are especially deployed in areas without established infrastructure. In [?], UAVs are used in such infrastructure-less areas, where their operations depend on both task offloading and trajectory planning. These UAVs are equipped with batteries and can access to charging facilities, allowing them to sustain their operations while navigating through complex environments to perform various Edge computing tasks. Similarly, EVs play a significant role in Edge computing systems. In [?], charging stations for EVs can be powered by both grid and renewable energy sources and they provide the possibility for EVs to discharge their energy, which can then be used to power other vehicles in need. While EVs in this scenario do not function directly as Edge infrastructure, they depend on a MEC system to coordinate the demands of charging and energy distribution among multiple vehicles.

Battery management in vehicular networks also involves strategies beyond traditional charging. In some cases, battery swapping is a viable alternative to reduce downtime and ensure continuous operation. For example, in the post-disaster scenarios described in [?], UAVs are deployed to provide ground-based devices with Edge computing services. These UAVs, equipped with batteries, are supported by battery swapping points located near or within the disaster area. The batteries in these swapping stations are recharged using renewable energy, enabling the UAVs to remain operational without extended delays for charging. Electric vehicles can also utilize battery switch stations as a means of maintaining energy levels. As described in [?], EVs that fall below a certain state of charge threshold must find a battery switch station. In this scenario, a centralized controller manages the coordination of EVs and battery switch stations through MEC servers. A reservation system ensures that EVs can access the necessary battery switch services, maintaining efficient energy distribution across the network.

Finally, some vehicular networks incorporate self-powered vehicles that use renewable energy to sustain their operations. In [?], smart vehicles equipped with solar panels and batteries demonstrate the potential for vehicles to rely on green energy sources. These vehicles are capable of being powered by solar energy, and when necessary, they can switch to fossil fuel-based energy to ensure continued operation. This scenario highlights the challenges of managing energy resources in a mobile network while also accounting for the placement of VNFs and traffic routing. The mobility of self-powered vehicles adds an extra layer of complexity to Edge computing orchestration, as these vehicles must dynamically adjust their energy use and task assignments based on real-time conditions.

In summary, batteries play a central role in enabling the functionality and autonomy of vehicles of an Edge vehicular network, whether through traditional charging, battery swapping, or self-sustaining renewable energy solutions. Mobile nodes, such as those found in vehicular networks, often face challenges in accessing the power grid and therefore rely heavily on batteries. Due to their mobility, their energy consumption is also influenced by external factors (what we previously referred to as device-related energy consuming activities), as highlighted in [?, ?].

2.2.2.5 On-spot energy strategies

Several studies adopt what can be described as an *on-spot* energy strategies, where Edge nodes operate directly from real-time renewable energy production without relying on any storage system. In [?], this strategy is applied to a battery-less green Edge gaming scenario compliant with the ETSI MEC standard. Energy is consumed immediately as it is produced, and the authors intentionally remove batteries to simplify the system design. However, this choice inherently reduces operational flexibility, as no energy can be buffered for future use. Job deployment and migration must adapt dynamically to instantaneous renewable availability.

A similar on-spot energy approach is adopted in [?], which investigates an IoV environment where vehicles interact with a mobile Edge host powered solely by renewable sources. Here, the absence of batteries is motivated by practical concerns, such as high cost, maintenance complexity and limited battery lifespan. As a consequence, the system must fall back to the power grid whenever renewable energy production dips below demand, making the orchestration tightly coupled to short-term energy fluctuations.

Overall, these studies illustrate a deliberate on-spot renewable energy strategy in Edge computing : avoiding batteries to reduce system cost and complexity while accepting the constraints of real-time energy use. Such designs face challenges related to continuity and reliability, as the availability of Edge resources depends entirely on the immediate renewable supply (or grid access when available). These approaches are rather rare, and contrast with the majority of scenarios in the literature, where renewable energy is combined with energy storage.

2.2.3 Renewable energy management

The integration of renewable energy sources into traditional Edge computing systems, derived from Cloud systems, which have historically relied on the power grid, presents both challenges and opportunities. This shift necessitates a comprehensive study of energy management strategies that can effectively handle the unpredictability of renewable sources. By considering a diverse energy inputs, from green to conventional grid power, we can explore approaches to optimize energy utilization, enhance system resilience, and reduce environmental impact in Edge computing environments.

2.2.3.1 Power prediction

Renewable energy sources, while intermittent, exhibit recognizable patterns and behaviors that we can make advantage of. By accurately forecasting available power, systems can optimize service orchestration to minimize energy waste and ensure continuity of operations.

For instance, in [?], a power management solution making use of a Lenovo Smart Power Agent uses platform telemetry and sensors to predict energy availability. This prediction is made using a pre-trained machine learning model, which continuously monitors system data. The model's output provides interfaces for external platforms like Prometheus and ODO, allowing for integrated energy management across various systems.

In other scenarios, energy harvesting predictions play a crucial role in managing renewable sources. For example, [?] estimates the energy harvested from photovoltaic systems within a defined time window. Future power values from PV systems are modeled as a Gaussian random variable with a given average value and standard deviation. A receding horizon technique is applied to solve the scheduling problem over that window. This approach allows the system to optimize task execution based on expected energy availability, ensuring that jobs are allocated when renewable energy is most abundant.

Solar energy forecasting, as demonstrated in [?], relies on meteorological data to predict the solar energy supply. Factors like sunlight intensity, surface area, time window and energy conversion rates are considered, though solar radiation remains the most critical variable, as the most of uncertainty comes from it. This forecast feeds into an energy-aware task scheduling system, enabling efficient energy distribution based on expected solar energy input. The authors of [?] also consider meteorological data to predict the light demand and power demand for solar-powered street lamps using quantile regression. In space-based scenarios such as [?], sunlight availability is more predictable since it does not depend on meteorological models. However, these systems must still account for various parameters including seasonal variations and constellation characteristics such as orbital inclination and altitude.

In these examples, power prediction serves as a tool for ensuring energy efficiency in Edge computing systems, allowing for smarter energy allocation and improved system reliability. They also enable a more proactive approach to orchestration processes, such as service migration [?] or task offloading [?].

2.2.3.2 Hybrid sources

In Edge computing, hybrid energy systems that combine renewable sources and the power grid are often being adopted to address the fluctuating availability of green energy and optimize energy costs. This approach ensures that Edge nodes maintain continuous operation by switching between renewable energy when available and the power grid when necessary. The motivation behind hybrid systems lies in the varying supply of renewable energy and the need for cost efficiency and reduced carbon emissions.

One of the key reasons for integrating the power grid alongside renewable sources is the time-varying nature of green energy availability. For instance, in [?], some Edge nodes access the grid to complement renewable energy sources when needed, considering the varying costs associated with different energy supplies. Similarly, [?] explores how the use of the power grid is triggered when renewable energy is insufficient, focusing on minimizing the cost of electricity drawn from the grid. This strategy is also supported by [?], where the goal is to minimize network-wide energy consumption from the grid when green energy production falls short. Hybrid systems also account for the location-based variation in energy prices and carbon footprints, as seen in [?]. In this case, Edge nodes experience differing energy prices and environmental impacts based on their geographic locations, allowing for more control over energy usage. Edge nodes can dynamically switch between renewable and grid power based on the cost and carbon intensity of energy sources, ensuring both economic and environmental efficiency.

In some scenarios, when renewable energy runs out, systems must rely entirely on the grid for continued operation. [?] discusses the transfer of green energy between nodes until it is fully depleted, after which brown energy from the grid is used. The focus in this case is on minimizing the cost associated with brown energy. In [?], electric network constraints and carbon emissions are considered through an energy sharing system. Similarly, in [?], a multi-tier Edge level is designed that can decide the amount of green energy to use in the system, and in [?], devices can receive green power via wireless power transfer from the edge when needed. Energy price management is also a consideration in hybrid systems. For example, in [?], macro base stations control energy flow among smaller base stations, taking into account the price differences between green and grid energy, while in [?], the authors account for the cost of energy based on a ratio of green energy supply. Moreover, [?] proposes a system where Edge nodes purchase and sell energy from and to the power grid, aiming for a system balance between energy consumption and generation. The topic of energy prices is further discussed in Section ??.

While many solutions focus on integrating the power grid with renewable energy to ensure a stable energy supply, the reverse approach is also viable. In fact, improving grid load efficiency by incorporating green energy is crucial. As explored in [?], renewable energy can play a key role in relieving the power grid, enhancing overall system resilience and reducing reliance on traditional energy sources.

In summary, hybrid energy systems provide a flexible and cost-efficient solution for Edge computing by blending renewable sources with grid power. This approach addresses the variability in green energy availability and helps reduce overall carbon emissions while ensuring that Edge nodes can operate continuously, regardless of energy source limitations.

2.2.3.3 Energy scheduling

We refer to *energy scheduling* as the strategic management and allocation of energy resources in real time or in advance, as a parallelism to task scheduling in service orchestration. First, this process involves deciding when to use energy, such as storing it for later demand. In [?], the concept of green energy scheduling allows for the use of only a fraction of the produced renewable energy in each time slot. The system can store any unused energy in a battery for future use, ensuring that surplus energy is not wasted. This method provides flexibility in energy consumption by enabling Edge nodes to reserve green energy for periods of high demand or when renewable energy generation is low, thereby improving overall energy efficiency. Energy scheduling becomes pertinent when operating under limited energy constraints. In [?], an energy management module monitors devices and allocates energy resources to balance consumption against task performance. This allocation uses a weighted function that considers task priorities, network conditions and remaining battery capacity.

Then, another approach to green energy scheduling is presented in [?], where the process is modeled as a graph. Each node in the system generates a certain amount of green energy during each time slot, and this energy can be transmitted between nodes through edges. This graph-based scheduling method not only manages energy distribution but also supports virtual machine (VM) migration, where the location of VMs can be adjusted based on available energy resources across nodes. By efficiently managing energy flow and VM placement, this approach ensures that renewable energy is optimally distributed throughout the edge network, reducing energy waste and enhancing the overall sustainability of the system.

The goal of energy scheduling is to optimize the use of renewable energy, by reducing waste and align energy consumption with availability, particularly in dynamic environments like Edge computing systems powered by renewable energy.

2.2.3.4 Energy pricing

In Edge computing systems, energy costs can vary based on the energy source used and different studies propose diverse strategies for optimizing these costs. A recurring theme is the cost disparity between green energy and brown energy.

Several studies, including [?], [?] and [?], treat green energy as free. In these cases, the focus is on minimizing the use of brown energy, which incurs a non-negligible cost. For example, in [?], the cost of brown energy is balanced with revenue generated from job execution, alongside other factors like deployment and migration costs. Similarly, [?] and [?] aim at minimizing system costs by using renewable energy as much as possible, only purchasing energy from the grid when necessary. In this cases, the goal is to maximize efficiency and cost savings by relying primarily on green energy.

Other studies introduce different price structures for green and brown energy. For instance, [?] assigns lower cost coefficients to green energy compared to brown energy, with the objective to prioritize renewable sources. This differential pricing is also seen in [?], where the price for brown energy is higher, and the system's objective is to minimize total energy costs by optimizing the use of green energy across nodes. In [?], a similar cost structure is applied, with backup energy sources, such as diesel generators, carrying a significantly higher cost, encouraging systems to rely on solar power whenever possible. Similarly, [?] examines energy pricing through both the price and greenness of the energy supply. This study considers the cost of processing microservices based on the type of energy used, with green energy being more desirable not only because of its lower price but also due to its environmental benefits. The system's objective is to minimize both computational and bandwidth provisioning costs while accounting for energy sources. Lastly, [?] integrates energy costs into a weighted sum of overall system consumption. In scenarios where harvested renewable energy is insufficient, the system purchases grid energy, which adds to the operational cost. This approach seeks to minimize energy expenses by dynamically balancing renewable and grid energy use, ensuring that the system remains cost-effective while maintaining its energy demands.

Another approach to energy costs is proposed in [?], where small base stations can trade surplus energy through a peer-to-peer energy trading market. In this scenario, energy is treated as a commodity, with base stations acting as either buyers or sellers, depending on their energy needs. This trading system is integrated into an OPEX minimization problem, where the system seeks to reduce costs by dynamically trading energy between nodes. Similarly, [?] presents a scenario where each Edge node can either purchase energy from the power grid or sell excess energy back to it. This energy trading process is managed through an energy buffer that balances harvested energy, energy consumption and stored energy. However, this model does not take monetary costs into account, assuming that purchased and sold energy have the same price, which simplifies the energy trading process.

In all these approaches, the common objective is to minimize total energy costs by prioritizing the use of green energy, which is either free or significantly cheaper, while reducing reliance on brown energy to maximize cost efficiency in Edge computing environments. Overall, energy pricing in Edge computing systems is closely related to the use of hybrid energy sources, where both green and brown energy are used to balance costs and ensure continuous operation. By effectively managing energy costs and trading mechanisms, these systems can optimize energy use, improve resilience, and reduce their carbon footprint.

2.3 Case studies

This section examines three representative application domains where renewable energy-enabled Edge computing faces distinct challenges and opportunities. These case studies illustrate the practical complexities of implementing the orchestration strategies discussed in our survey. We identified three recurrent representative scenarios in the literature, and we selected them as case studies.

2.3.1 Autonomous vehicles

The autonomous vehicle ecosystem presents unique challenges for renewable energy-enabled Edge orchestration, particularly in device discovery, scalability, and security management.

— *Device discovery and privacy challenges :* Traditional device discovery mechanisms become problematic in vehicular environments due to privacy and security concerns. Vehicles cannot safely broadcast their exact locations or computational capabilities to all nearby devices, as this information could be exploited for tracking or targeted attacks [?].

— *Hybrid energy management :* Charging stations equipped with hybrid energy sources, combining solar panels, wind generation, and grid connectivity show the complexity of managing heterogeneous renewable energy. As demonstrated in [?], these stations must dynamically balance energy from multiple sources while coordinating with vehicle charging demands and computational service provisioning.

— *Battery management and longevity :* Vehicle-integrated Edge computing faces the dual challenge of managing battery resources for both mobility and computational services. Effective battery management strategies must consider charging cycles, thermal management, and degradation patterns to maximize battery lifespan while ensuring computational availability. This creates complex trade-offs between service quality, energy efficiency, and long-term operational costs, as expressed in works like [?, ?].
— *Infrastructure-less scenarios :* The research in [?] demonstrates how UAVs can provide computational services in areas lacking traditional infrastructure. Solar-powered UAVs can serve as mobile Edge nodes, but their energy-constrained operation requires careful trajectory planning and service scheduling to balance energy consumption with service delivery.
— *Scalability implications :* As autonomous vehicle adoption increases, the number of mobile Edge nodes could grow from hundreds to thousands within urban areas, creating unprecedented scalability challenges for energy-aware orchestration systems [?]. Current centralized coordination approaches become impractical at this scale, necessitating distributed orchestration mechanisms that can operate with limited inter-vehicle communication.

2.3.2 Smart agriculture

Rural agricultural environments present distinct challenges for renewable energy integration, primarily due to limited grid infrastructure and the need for specialized agricultural applications.

— *Grid connectivity limitations :* Rural Edge deployments frequently lack reliable grid connectivity, making renewable energy integration not just beneficial but essential for operational sustainability. Unlike urban deployments where grid power serves as backup, agricultural Edge systems must be designed as energy-independent entities capable of sustained operation during extended periods of poor renewable energy generation as in [?, ?].
— *Specialized agricultural applications :* The work in [?] demonstrates practical renewable energy integration through solar-powered insecticidal lamps that combine pest control with Edge computing capabilities. These devices use predictive algorithms to optimize pest elimination timing while managing limited solar energy resources. The research presented in [?] explores the deployment of solar-powered Edge devices for crop monitoring and disease prediction applications. These systems must operate continuously during critical growing periods, requiring robust energy storage solutions to handle cloudy weather or seasonal variations in solar availability.
— *Seasonal variations :* Agricultural deployments face significant seasonal va-

riations in both energy generation and computational demands. Solar generation peaks during summer months when crops require intensive monitoring, but spring and fall periods may combine high computational demands with reduced solar availability. Seasonal variations in sunlit ratios have already been analyzed in [?], and similar techniques could be applied to smart agriculture. Wind patterns in agricultural areas also vary seasonally, requiring adaptive energy management strategies that account for both annual and daily variation patterns.

— *Remote maintenance challenges :* The distributed nature of agricultural Edge deployments makes maintenance and system updates particularly challenging. Renewable energy components require regular cleaning and maintenance, but the remote location of many agricultural Edge nodes makes frequent service visits economically costly [?]. This necessitates robust, self-maintaining systems with redundant energy sources, and remote diagnostic capabilities.

2.3.3 Smart cities

Urban environments present complex challenges for renewable energy-powered Edge computing due to high population density, diverse energy requirements, and significant geographic variations in renewable energy generation.

— *Urban renewable energy constraints :* Cities face unique challenges in renewable energy deployment due to limited space, building shadows, and air quality issues that affect solar panel efficiency. Urban wind patterns are highly variable due to building effects, making small-scale wind generation less predictable than in rural environments. Solar panels have also similar problems as expressed in [?].

— *Geographic variations :* Different cities face dramatically different renewable energy potential [?].

— *Grid integration complexity :* Smart city deployments benefit from existing electrical grid infrastructure but face complexity in integrating renewable energy into their current grid. The work in [?] demonstrates approaches that consider renewable energy integration into the broader power grid. This integration requires sophisticated coordination between Edge computing energy demands, renewable energy generation and grid stability requirements.

— *Multi-tenant coordination :* Urban deployments involve multiple actors including city governments, utility companies, private service providers, and citizens. Each actor has different priorities regarding energy efficiency, service quality, and cost optimization [?]. Effective orchestration must balance these competing interests while maintaining system-wide efficiency and sustainability goals.

2.4 Conclusion

In this chapter, we compiled and categorized a comprehensive set of existing service orchestration strategies that decrease the energy consumption in Edge computing, as well as novel strategies introduced by integrating renewable energy sources into the orchestration process to understand how renewable energy integration changes orchestration. This classification is summarized with Fig. ??. We also provided three case studies for future investigations in this field.

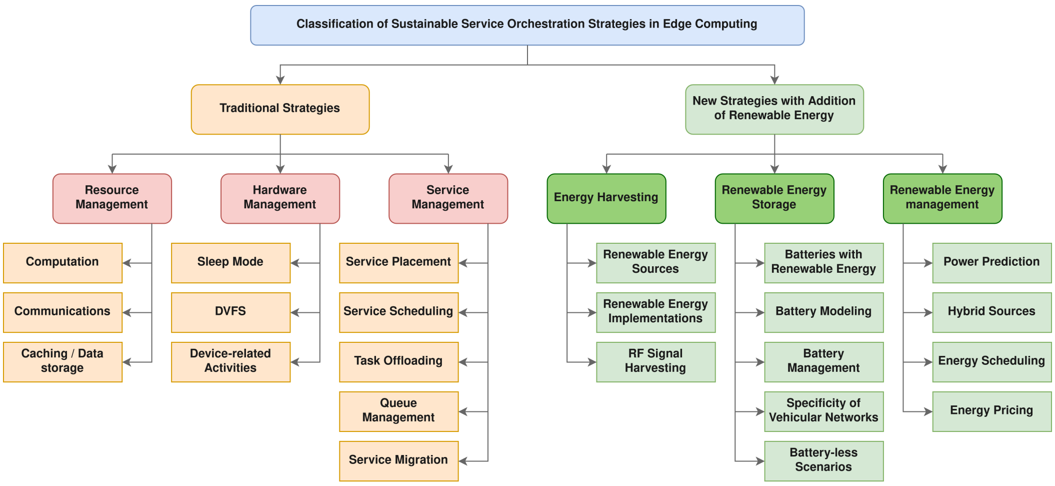

FIGURE 2.1 – Classification of sustainable service orchestration strategies in Edge computing.

From this study, we identified research gaps and provide future directions :

1. *Heterogeneity of resources :* Edge systems often have heterogeneous resources with varying capabilities, energy consumption patterns and performance characteristics that need accurate resource models to capture the disparities between devices and enable effective service orchestration decisions. Most articles in the literature only consider one type of green energy in their work, whereas different sources could be provided to supply an Edge node.

2. *Implementation :* Currently, there are few real-world implementations of Edge systems, and even fewer that incorporate renewable energy sources. To truly validate the effectiveness of these approaches, it is crucial to conduct practical implementations and gather empirical evidence.

3. *Energy storage :* Just as there are diverse types of green energy sources, there are also multiple methods of energy storage that could be integrated into Edge systems using renewable energy as mentioned in [?].

4. *Scalability :* As the number of services and devices might vary in a system, managing resource and ensuring scalability can be difficult. Indeed, most works consider static infrastructures with a constant number of Edge nodes.

Scalable orchestration mechanisms should be implemented to handle the dynamic applications and infrastructure topology.

5. *Device discovery :* The dynamic discovery of devices that could potentially host greener energy resources presents a unique challenge in this context. While traditional device discovery in Edge computing primarily focuses on identifying nearby computational resources based on factors such as processing power, storage capacity and network capabilities [**?**], the discovery of energy-efficient resources adds a new layer of complexity to this process. This complexity arises from the need to consider not only the device's computational capabilities but also its energy profile and potential access to renewable or more efficient energy sources. The discovery mechanism must be flexible enough to accommodate several scenarios. For instance, a static device could make use of nearby mobile devices with energy resources, such as electric vehicles or portable solar panels. Also, it might be a mobile device that can take advantage of static energy sources in its vicinity, such as charging stations or building-integrated solar panels.
6. *AI-driven predictive analysis :* The feasibility of integrating AI-driven predictive analytics for renewable energy-based orchestration presents both opportunities and challenges. However, these models require continuous training data collection from multiple sensors (weather, energy production, consumption), like in [**?**, **?**] and robust handling of prediction uncertainties that current orchestration algorithms inadequately address.
7. *Cost of renewable energy :* Current research insufficiently addresses the economic realities of renewable energy deployment. Furthermore, these systems have a limited lifespan (especially batteries) and a maintenance cost must be considered as in [**?**]. An alternative is to adopt *on-spot* energy strategies, as in [**?**, **?**], which rely directly on real-time renewable energy but must deal with its inherent variability.
8. *Geographic limitations :* Solar-powered Edge systems face significant regional variations in power production [**?**]. Similarly, wind-based systems are viable only in locations with consistent wind patterns, limiting deployment to specific geographic regions.
9. *Complexity of optimization :* Multi-objective optimizations problems are complex, especially when dealing with large-scale distributed systems with numerous variables and constraints. Some advanced optimization techniques can be employed, like machine learning [**?**], multi-agent systems [**?**] or evolutionary algorithms [**?**] to find near-optimal solutions.
10. *Stakeholders coordination :* In many deployments, energy production and storage, managed by building owners, facility managers or energy providers, and application orchestration, managed by Edge operators or service providers, are handled by different entities with limited information exchange. This separation can make it difficult to coordinate traditional and renewable

energy strategies, especially when energy-availability data, storage status, or grid-use policies are not accessible to orchestration systems. Prior surveys on energy-aware Edge deployment highlight that organizational and governance issues, such as heterogeneous ownership, conflicting objectives and information asymmetry, remain major open challenges in moving toward green orchestration frameworks [**?**]. Future work should therefore explore mechanisms for cross-stakeholder cooperation, shared energy metadata and energy-aware SLAs to enable coordinated and practical renewable energy-enabled orchestration.

11. *Security and privacy concerns :* Introducing new mechanisms for energy efficiency and energy sources may introduce new vulnerabilities or affect the security and privacy of the system.

Research directions 1, 6, and 7 are tackled in Chapter **??** and 2 in Chapter **??**.

Chapitre 3

Sustainable service placement & service placement reconfiguration

Sommaire

3.1 Concepts, Architecture & Models

This section introduces the fundamental elements of our approach. We begin by detailing the key concepts used throughout our work. We then present an overview of the proposed architecture, illustrated in Fig. **??**, highlighting its main modules and the relationships between infrastructure, services, and orchestration mechanisms. Building on this structural description, we detail the models that characterize the infrastructure and its energy behavior, including the abstraction of computational resources, workload representation, and power dynamics. To facilitate readability and ensure precision in the mathematical formulations that follow, we provide a comprehensive table (Table **??**) of notations summarizing the symbols and variables used across this chapter.

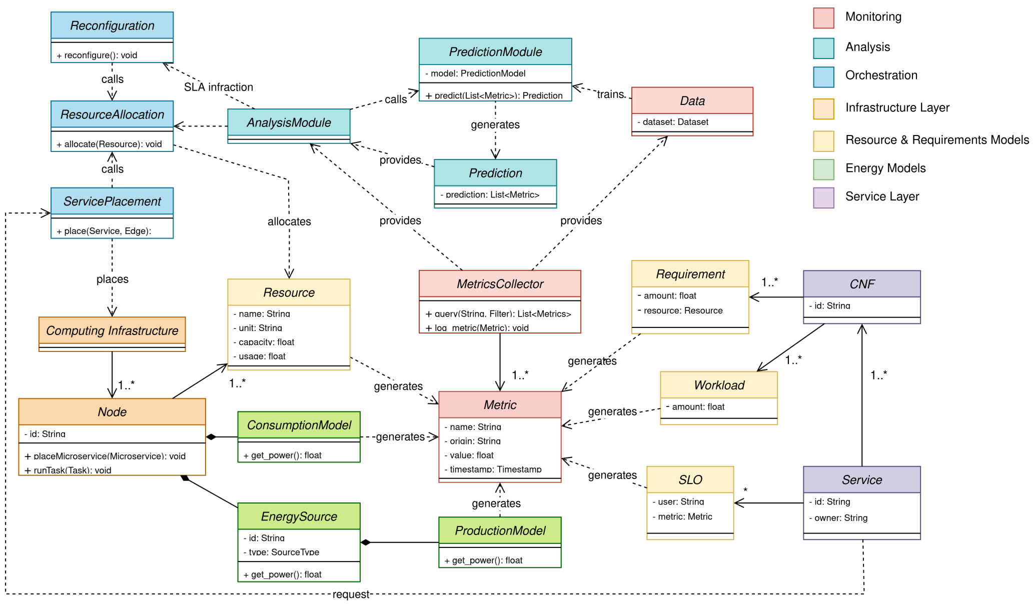

FIGURE 3.1 – Overview of the system architecture

3.1.1 Service, Infrastructure & Resources

Our considered system adopts a container-based architecture composed of three main elements :

- *Services* : Logical groupings of CNFs. Each service $s \in S$ is defined by a set of CNFs denoted as F_s.
- *CNFs* : Individual containerized network functions.
- *Nodes* : Physical infrastructure elements with finite computational resources.

We primarily focus on processing (CPU) and memory (RAM) as key resources. Each node N is defined by its allocatable resources $\{CPU_n^{alloc}, RAM_n^{alloc}\}$. For each CNF $f \in F$, the resource requirements are given as $f = \{CPU_f^{req}, RAM_f^{req}\}$. Regarding CPU, we distinguish between allocated resources and the resources currently in use. Specifically, each CNF f has a time-dependent CPU workload at time t, denoted by $W_f^{CPU}(t)$ and defined in (**??**).

$$CPU_n^{\%}(t) = \frac{\sum_{f \in F_n} W_f^{CPU}(t)}{CPU_n^{alloc}}. \tag{3.1}$$

3.1.2 Energy

An energy consumption model is associated to each node n and a renewable energy generation model is considered when available on the node.

Notation	Meaning
N	Set of nodes
S	Set of services
F	Set of CNFs
F_s	Set of CNFs that compose service s
F_n	Set of CNFs running on node n
CPU_f^{req}	Required CPU of CNF f
RAM_f^{req}	Required RAM of CNF f
W_f^{CPU}	CPU workload of CNF f
CPU_n^{alloc}	CPU capacity of node n
RAM_n^{alloc}	RAM capacity of node n
CPU_n	CPU usage of node n (in cores)
$CPU_n^{\%}$	CPU usage of node n (in percent)
PM_n	Power consumption model of node n
P_n	Power consumption of node n
P_n^{green}	Renewable power generation of node n
P_n^{brown}	Non-renewable power consumption of node n
P_n^{wasted}	Wasted power of node n
X	Binary variable for CNF placement
M	Binary variable for CNF migration
B^{CPU}	Binary variable for CPU over-allocation
B^{RAM}	Binary variable for RAM over-allocation
$\widetilde{W}_f^{CPU}$	CPU workload prediction for CNF f
$\widetilde{P}_n^{green}$	Renewable power generation prediction for node n
$\widetilde{G}_n$	Green ratio of node n
$\widetilde{G}_s$	Green ratio of service s
$\widetilde{G}_s^{valid}$	Green ratio constraint satisfaction of service s
G_s^c	Green ratio threshold of service s
SCR_s	Service concentration ratio of service s
m_f	Migration cost of CNF f

TABLE 3.1 – Notation table

3.1.2.1 Energy consumption model

We define the power consumption P_n of a node n as a function of its CPU utilization. In our work, the adopted model PM_n (**??**) is a polynomial of degree m with coefficients a_i obtained via polynomial regression from experimental power data on the platform Grid'5000 [**?**] as displayed in Fig. **??**.

$$PM_n(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_m x^m. \tag{3.2}$$

To create a model correlating CPU usage with power consumption, we designed an experimental method. We executed a CPU-intensive script on a node, simultaneously measuring both CPU utilization and power consumption.

To vary the CPU load, we used `stress-ng`, a Linux-based stress-testing tool. This tool allows precise control over the number of CPU cores used, their load percentage, and the duration of each workload.

CPU usage was measured using the `psutil` Python library (`cpu_percent` function), which provides real-time CPU utilization metrics. On Linux, `psutil` can use the `/proc` filesystem and make calls to `sysinfo`, `getloadavg`, etc.

Power consumption was retrieved using the Grid'5000 API. The platform uses wattmeters to measure node power draw at one-second intervals. All metrics were retrieved post-experiment using the experiment's timestamps.

Our script follows a three-phase approach :

1. *Idle phase* : 30 seconds without load.
2. *Fixed CPU load test* : Sequential stress tests at 10% increments (10%, 20%, ..., 100%) for one minute each, with a 5 seconds cool down, applied across all cores.
3. *Mixed workload test* : Randomized workloads with variable core usage, load percentages, and durations.

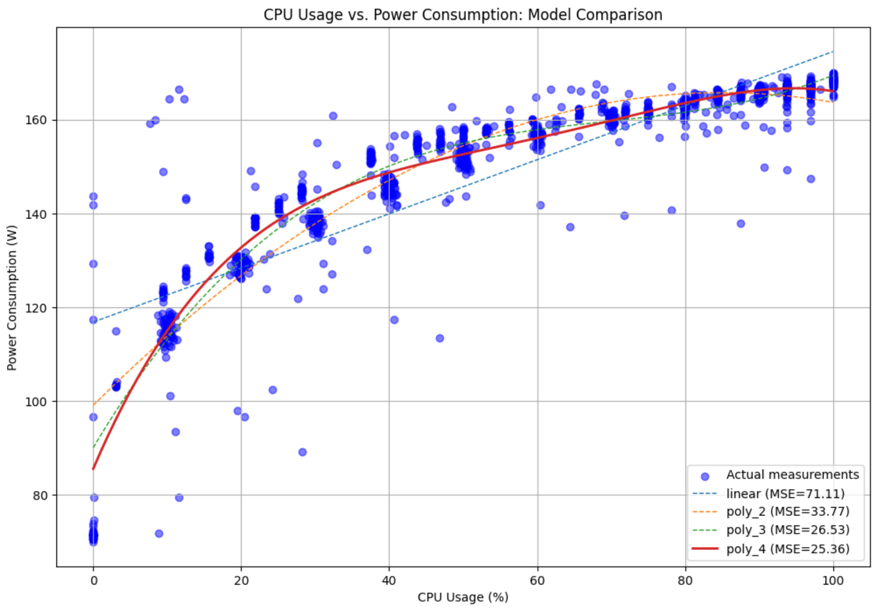

FIGURE 3.2 – Power consumption models obtained via polynomial regression (up to degree 4) from Grid'5000 experiments.

3.1.2.2 Energy generation model

When renewable generation is available on a node n, we define the green power output as a time-varying function $P_n^{green}(t)$. The residual power drawn from the electrical grid (that we call brown power in our work) is then defined by (**??**).

$$P_n^{brown}(t) = \begin{cases} P_n(t) - P_n^{green}(t) & \text{if } P_n(t) > P_n^{green}(t), \\ 0 & \text{otherwise.} \end{cases} \tag{3.3}$$

In our model, renewable generation can either come from solar or wind sources.

Solar power generation is modeled using a normal distribution-based irradiance profile, estimated via Kernel Density Estimation (KDE) as used in [**?**]. The instantaneous solar output is then $P_n^{solar}(t) = \eta_n \, S_n \, I_n(t)$, where η_n is the panel efficiency, S_n the panel surface area, and $I_n(t)$ the irradiance at time t. The process is detailed in Fig. **??**.

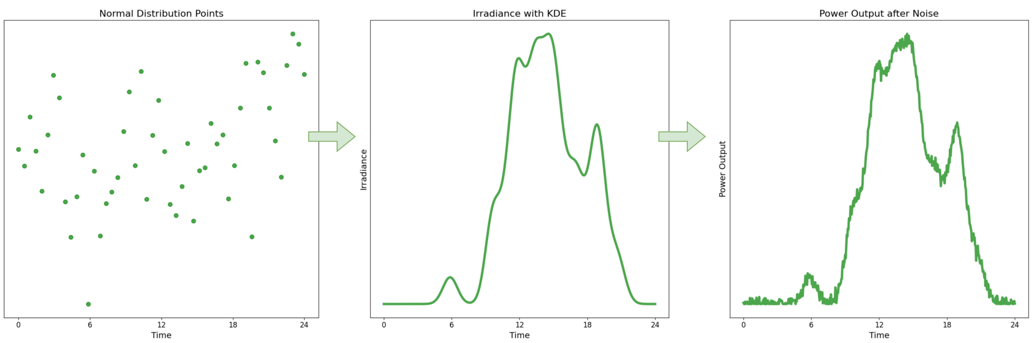

FIGURE 3.3 – Step-by-step visualization of solar power generation from a normal distribution using KDE.

Wind power generation model is derived from wind speed generation using a Weibull distribution based on the work in [**?**]. The final power output can be computed with (**??**), where P_n^{rated} is the rated power output, $v_n(t)$ the wind speed, and $v_n^{cut\_in}$, v_n^{rated}, $v_n^{cut\_out}$ the cut-in, rated, and cut-out speeds, respectively. This is also explained by Fig. **??**. For both renewable generation models, we add a Gaussian noise to account for natural variability.

$$P_n^{wind}(t) = \begin{cases} P_n^{rated} \cdot \dfrac{v_n(t) - v_n^{cut\_in}}{v_n^{rated} - v_n^{cut\_in}} & \text{if } v_n^{cut\_in} \leq v_n(t) < v_n^{rated} \\ P_n^{rated} & \text{if } v_n^{rated} \leq v_n(t) \leq v_n^{cut\_out} \\ 0 & \text{otherwise.} \end{cases} \tag{3.4}$$

This stochastic approach enables us to evaluate our methodology across diverse power generation models, eliminating reliance on a fixed dataset. However, for prac-

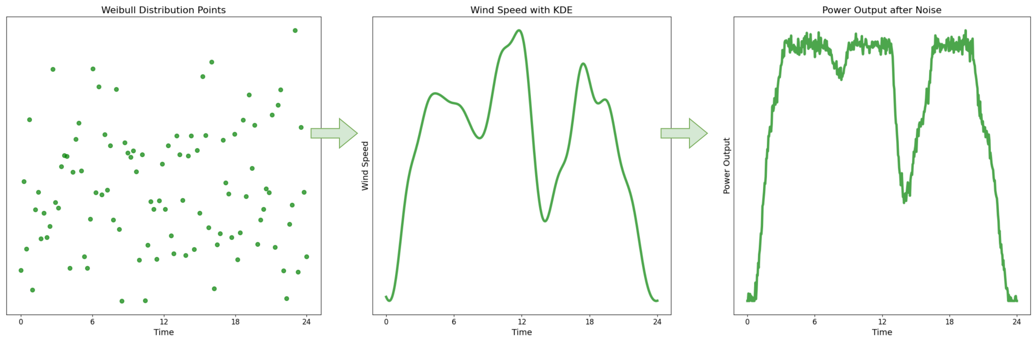

FIGURE 3.4 – Step-by-step visualization of wind power generation from a Weibull distribution using KDE.

tical applications, using real-world measurements may be more appropriate.

3.1.3 Monitoring

Monitoring ensures continuous visibility into the state and performance of both physical and virtual entities. It encompasses a wide range of components, from physical nodes to virtual services like our CNFs. Energy-related infrastructure, including power consumption and renewable energy generation, is also closely tracked.

The metrics collected are diverse and purpose-driven. Physical resources like CPU and RAM are monitored alongside virtual resource utilization. Energy metrics, including consumption and generation from renewable sources, are also monitored. Additionally, the system tracks the status and performance of nodes and services, capturing health indicators, uptime, and response times.

The data gathered serves multiple purposes. It provides real-time insights for orchestrators, enabling them to make informed, dynamic decisions. It also acts as a valuable dataset for training machine learning models, supporting predictive analytics and long-term optimization. Finally, this data is transmitted to analysis modules, where it is processed to uncover trends, anomalies, and opportunities for improvement.

3.1.4 Analysis

The analysis module transforms raw monitored data into usable intelligence. By aggregating and correlating metrics across the system, it identifies patterns, detects anomalies, and highlights lack of resources. This processing is not just reactive, it can also be used to make predictions, such as energy availability based on renewable generation trends or anticipating workload demands during peak periods.

These insights directly influence the orchestrator's decisions. For example, predictions about energy shortages or workload spikes can trigger proactive adjustments, ensuring the system remains efficient, reliable, and aligned with operational goals.

3.1.5 Orchestration

Orchestration is the decision-making engine of the system, responsible for dynamically allocating resources and managing services in accordance with predefined policies and real-time insights. It ensures that services are placed optimally, balancing factors such as performance, energy efficiency, and cost. Resource allocation is not static, it evolves based on demand, availability, and policy constraints. A key function of orchestration is the dynamic migration of CNFs.

In our problem, we consider an Edge service orchestration problem where services, composed of multiple CNFs, must be deployed on a set of nodes with limited resources and subject to unpredictable node departures. The objective is to find optimal *service placement* and *service placement reconfiguration* strategies that makes the best use of available energy resources while ensuring service continuity.

3.1.5.1 Service Placement

Service placement orchestrator assigns each CNF to a physical node while satisfying resource constraints and optimizing multiple objectives. We define a binary decision variable $X(n, f, t) \in \{0, 1\}$ that equals 1 when a CNF f is placed on a node n. Each CNF must be placed on exactly one node, and the total resource requirements of CNFs placed on a node must not exceed its allocatable resources.

3.1.5.2 Service Placement Reconfiguration

We then define service placement reconfiguration as the process of migrating CNFs from a node to another in response to infrastructure changes, such as energy variations or node departures. We use the same variable $X(n, f, t)$ and introduce a new migration variable $M(n, f, t) \in \{0, 1\}$ that equals 1 if CNF f is migrated to node n at time t. The reconfiguration process must also satisfy the same resource constraints.

3.2 Optimization Metrics

We define main optimization metrics for our service placement and service placement reconfiguration problems. The placement problem is only considered at $t+1$, just after the placement, while the reconfiguration problem can either be considered at $t + 1$ or $t + h$ for the reactive and proactive reconfigurations respectively. To generalize the problem, we consider our metrics at a time $t + h$, where $h \in [1, H]$, with H our prediction horizon.

3.2.1 Overall power consumption

We compute the CPU usage of each node at time $t + h$ as in (**??**) and (**??**), from the prediction of CNF f workload $\widetilde{W}_f^{CPU}$. Then, in (**??**), we derive the power consumption using the power model of the node.

$$\widetilde{CPU}_n(t+h) = \sum_{f \in F} X(n,f,t+h) \cdot \widetilde{W}_f^{CPU}(t+h). \tag{3.5}$$

$$\widetilde{CPU}_n^{\%}(t+h) = \min\left(1, \frac{\widetilde{CPU}_n(t+h)}{CPU_n^{alloc}}\right). \tag{3.6}$$

$$\widetilde{P}_n(t+h) = PM_n\left(\widetilde{CPU}_n^{\%}(t+h)\right). \tag{3.7}$$

3.2.2 Brown power consumption

The brown power consumption is then obtained in (**??**) by subtracting the prediction of power generation $\widetilde{P}_n^{green}$ to the estimated power consumption.

$$\widetilde{P}_n^{brown}(t+h) = \begin{cases} \widetilde{P}_n(t+h) - \widetilde{P}_n^{green}(t+h) & \text{if } \widetilde{P}_n(t+h) > \widetilde{P}_n^{green}(t+h), \\ 0 & \text{otherwise.} \end{cases} \tag{3.8}$$

3.2.3 Wasted power

Similarly, the wasted power can be obtained using (**??**) by subtracting the estimated power consumption to the prediction of power generation $\widetilde{P}_n^{green}$.

$$\widetilde{P}_n^{wasted}(t+h) = \begin{cases} \widetilde{P}_n^{green}(t+h) - \widetilde{P}_n(t+h) & \text{if } \widetilde{P}_n(t+h) < \widetilde{P}_n^{green}(t+h), \\ 0 & \text{otherwise.} \end{cases} \tag{3.9}$$

3.2.4 Service green ratio

From both the prediction of green power generation $\widetilde{P}_n^{green}$ and our estimation of power consumption $\widetilde{P}_n$ in (**??**), we can compute the *green ratio* of a node n at time $t+h$ as in (**??**), that represents the percentage of the energy consumed on the node that is green at time slot $t+h$. From this ratio, we define the green ratio of a service s as (**??**), assuming that the green ratio of a CNF is the ratio of the node it is placed on. The satisfaction of the green ratio constraint for a service is then defined in (**??**), where G_s^c is the minimum required green ratio for service s. Since we aim to avoid interruptions in our system, defining an acceptable threshold of green energy usage for our services is an alternative to purely maximize the green usage of our infrastructure. This constraint could be negotiated between the service provider and the infrastructure provider, that orchestrate the wanted services.

$$\widetilde{G}_n(t+h) = \begin{cases} \frac{\widetilde{P}_n^{green}(t+h)}{\widetilde{P}_n(t+h)} & \text{if } \widetilde{P}_n^{green}(t+h) < \widetilde{P}_n(t+h) \\ 1 & \text{otherwise.} \end{cases} \tag{3.10}$$

$$\widetilde{G}_s(t+h) = \frac{1}{|F_s|} \sum_{f \in F_s} \sum_{n \in N} X(n,f,t+h) \cdot \widetilde{G}_n(t+h). \tag{3.11}$$

$$\widetilde{G}_s^{valid}(t+h) = \begin{cases} 1 & \text{if } \widetilde{G}_s(t+h) \geq G_s^c, \\ 0 & \text{otherwise.} \end{cases} \tag{3.12}$$

3.2.5 Service concentration ratio

To minimize the impact of node departures on service continuity, we define the *service concentration ratio (SCR)* of a service s at time $t+h$ as in (**??**), where $|F_s|$ is the total number of CNFs of service s and $n_s(t)$ is the highest number of CNFs that on the same node at time t. The goal of this metric is to minimize the amount of CNFs on the same node in case of departure. $r \in \{0,1\}$ is a constant to define a threshold that is acceptable, e.g. 0.5 for a maximum of 50% of the service's CNFs on the same node. The idea behind this metric is to avoid clustering too much CNFs of the same service on the same node, that will cause an interruption of the service in case of departure. This objective is contradictory with the service green ration satisfaction, that would prioritize the choice of the "best" node, i.e. the greenest. Note that the SCR metric is different from load balancing, since we still can have large difference in the load of our nodes.

$$SCR_s(t+h) = \begin{cases} r \cdot \frac{|F_s|}{n_s(t+h)} & \text{if } n_s(t+h) > r \cdot |F_s| \\ 1 & \text{otherwise.} \end{cases} \tag{3.13}$$

3.2.6 Migration cost

Finally, we define the migration cost as in (**??**) for each migrated CNF $f \in \widehat{F}$, where $t_{init}(f)$ is the initialization time of a CNF f. Our logic behind this is that to preserve the continuity of the service, the original instance of the migrated CNF must continue to run until the new instance is initialized, leading to a surplus of consumption during this migration.

$$m_f(t+h) = \sum_{n \in N} M(n,f,t+h) \cdot t_{init}(f) \cdot CPU_f^{req}. \tag{3.14}$$

3.3 Algorithms

3.3.1 Heuristics

3.3.1.1 GREENING

Algorithme 1 : Greedy Placement

Input : N : Nodes, F_s : CNFs to place, X : Placement Matrix
Output : Placement mapping at $t+1$

$N^{(\tau)} \leftarrow sort(N, P_n^{green} - P_n)$;
for $f \in F_s$ **do**
    for $n \in N^{(\tau)}$ **do**
        if *n has enough resources for f* **then**
            $X(n, f, t+1) \leftarrow 1$;
            break;
    if $\sum_{n \in N} X(n, f, t+1) = 0$ **then**
        $X(0, f, t+1) \leftarrow 1$;
return X;

3.3.1.2 Lb-diversity

3.3.2 Genetic Algorithms

3.3.2.1 Chromosomes

3.3.2.2 Traditional Genetic Algorithm

3.3.2.3 NSGA-II

3.3.2.4 SP-GA

3.3.2.5 PSPR-GA

3.4 Metric Choice & Weight Influence

3.5 Balancing Sustainability & Robustness

3.6 Proactive Orchestration

3.7 Conclusion

Algorithme 2 : GREENING (migration)

```
Input : N : Nodes, F : Running CNFs, X : Placement Matrix, M :
        Migration Matrix
Output : Migration plan at t + 1
N^(τ) ← sort(N, P_n^green − P_n);
for n ∈ N^(τ) do
    if P_n^green(t) > P_n^green(t + 1) then
        candidates ← {f ∈ F,   X(n, f, t) = 1};
        while P_n(t + 1) > P_n^green(t + 1) do
            if candidates = ∅ then
                break;
            sorted_candidates ← sort(candidates, t_init(f) · CPU_f^req);
            for d ∈ N^(τ) do
                if d = n then
                    continue;
                if sorted_candidates[0] can be placed on d and new
                 P_n(t + 1) ≤ P_n^green(t + 1) then
                    X(d, f, t + 1) ← 1;
                    X(n, f, t + 1) ← 0;
                    M(d, f, t + 1) ← 1;
            Remove sorted_candidates[0] from candidates;
return M;
```

Algorithme 3 : Lb-diversity

Input : N : Nodes, F_s : CNFs to place, X : Placement Matrix
Output : Placement mapping at $t+1$

$\bar{F} \leftarrow$ sort F_s by sum of CPU and RAM ranking;
for $f \in \bar{F}$ **do**
    candidates $\leftarrow \emptyset$;
    $m \leftarrow -\infty$;
    for $n \in N$ **do**
        $n \leftarrow \sum_{k \in F} X(n, k, t)$;
        if $n < m$ **then**
            candidates $\leftarrow \{n\}$;
        else if $n = m$ **then**
            candidates $\leftarrow$ *candidates* $\cup \{n\}$;
    $\bar{N} \leftarrow$ sort *candidates* by sum of CPU and RAM usage;
    for $n \in \bar{N}$ **do**
        if f *can be placed on* n **then**
            $X(n, f, t+1) = 1$;
return X;

Algorithme 4 : Traditional Genetic Algorithm

Input : P : Population size, G : Generations, μ : Mutation probability, σ : Selection percentage
Output : Best individual

Initialize random *population*;
Evaluate *population*;
$g = 0$;
while *Termination criterion not met or* $g < G$ **do**
    $g = g + 1$;
    Select σ of best chromosomes of *population*;
    Perform crossover;
    Perform mutation with each gene having a probability of μ to mutate;
    Evaluate the new population;
return *Best individual of population*;

Algorithme 5 : Non-dominated Sorting Genetic Algorithm II (NSGA-II)

Input : P : Population size, G : Generations, μ : Mutation probability
Output : Best Pareto Front

Chapitre 4

From theory to practice : A more generalizable and applicable implementation with Open Source MANO

4.1 Challenges in Adapting Theoretical Models to Existing Standards

4.1.1 Multi-cluster orchestration

4.1.2 Abstraction gaps : containerized vs. virtualized Resources

4.1.3 The overlooked role of energy in NFV orchestrators

4.2 CNF placement and migration with OSM

4.2.1 Architectural overview of our system

4.2.2 OSM as a NFV MANO orchestrator

4.2.3 Deploying CNFs : Kubernetes, Helm & OSM Integration

4.2.4 Monitoring & optimization

4.3 Experimental validation

4.3.1 Experimental setup

4.3.2 Service placement experiments

4.3.3 Service migration experiments

4.4 Conclusion

Conclusion

Ce manuscrit de thèse rapporte

Annexe A

Exemple d'annexe

A.1 Exemple d'annexe